data.py > ...

```python
import pandas as pd

# Student Information
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covid_data = {
    "Country": [
        "India",
    ],

    "Confirmed_Cases": [
        44690738,
        103802702,
        37076053,
        22977474,
        39866718
    ],

    "Death_Cases": [
        530779,
        1123836,
        699276,
        399426,
        166176
    ],

    "Recovered_Cases": [
        44158339,
        101858395,
        36124991,
        22345821,
        39633299
    ],

    "Active_Cases": [
        1600,
        820471,
        251786,
        232227,
        67243
    ]
```

```
}

df = pd.DataFrame(covid_data)

print(df.head())

print(df.tail())

print(df.shape)

print(df.columns)

print(df.info())

print(df.isnull().sum())

print(df["Confirmed_Cases"].sum())

print(df["Death_Cases"].sum())

print(df["Recovered_Cases"].sum())

print(df["Active_Cases"].sum())

highest_cases = df[df["Confirmed_Cases"] == df["Confirmed_Cases"].max()]
print(highest_cases)

lowest_cases = df[df["Confirmed_Cases"] == df["Confirmed_Cases"].min()]
print(lowest_cases)

sorted_cases = df.sort_values(by="Confirmed_Cases", ascending=False)
print(sorted_cases)

top_deaths = df.sort_values(by="Death_Cases", ascending=False).head(3)
print(top_deaths)

print(df["Confirmed_Cases"].mean())

print(df["Active_Cases"].max())
```

```python
print(df["Death_Cases"].min())

df["Death_Rate"] = (df["Death_Cases"] / df["Confirmed_Cases"]) * 100
print(df[["Country", "Death_Rate"]])

df["Recovery_Rate"] = (df["Recovered_Cases"] / df["Confirmed_Cases"]) * 100
print(df[["Country", "Recovery_Rate"]])

df.to_csv("covid_analysis.csv", index=False)

print("CSV File Saved Successfully")
```

```
PS C:\Users\sham\Desktop\Pandas Library> & "C:\Program Files\Python314\python.ex
  Country  Confirmed_Cases  Death_Cases  Recovered_Cases  Active_Cases
0   India         44690738       530779         44158339          1600
1     USA        103802702      1123836        101858395        820471
2  Brazil         37076053       699276         36124991        251786
3  Russia         22977474       399426         22345821        232227
4  France         39866718       166176         39633299         67243
  Country  Confirmed_Cases  Death_Cases  Recovered_Cases  Active_Cases
0   India         44690738       530779         44158339          1600
1     USA        103802702      1123836        101858395        820471
2  Brazil         37076053       699276         36124991        251786
3  Russia         22977474       399426         22345821        232227
4  France         39866718       166176         39633299         67243
(5, 5)
Index(['Country', 'Confirmed_Cases', 'Death_Cases', 'Recovered_Cases',
       'Active_Cases'],
      dtype='str')
<class 'pandas.DataFrame'>
RangeIndex: 5 entries, 0 to 4
Data columns (total 5 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   Country          5 non-null      str
 1   Confirmed_Cases  5 non-null      int64
 2   Death_Cases      5 non-null      int64
 3   Recovered_Cases  5 non-null      int64
 4   Active_Cases     5 non-null      int64
dtypes: int64(4), str(1)
memory usage: 332.0 bytes
None
Country            0
Confirmed_Cases    0
Death_Cases        0
Recovered_Cases    0
Active_Cases       0
dtype: int64
248413685
2919493
244120845
1373327
  Country  Confirmed_Cases  Death_Cases  Recovered_Cases  Active_Cases
1     USA        103802702      1123836        101858395        820471
  Country  Confirmed_Cases  Death_Cases  Recovered_Cases  Active_Cases
3  Russia         22977474       399426         22345821        232227
  Country  Confirmed_Cases  Death_Cases  Recovered_Cases  Active_Cases
1     USA        103802702      1123836        101858395        820471
```

```
  Country  Confirmed_Cases  Death_Cases  Recovered_Cases  Active_Cases
3  Russia         22977474       399426         22345821        232227
  Country  Confirmed_Cases  Death_Cases  Recovered_Cases  Active_Cases
1     USA        103802702      1123836        101858395        820471
0   India         44690738       530779         44158339          1600
4  France         39866718       166176         39633299         67243
2  Brazil         37076053       699276         36124991        251786
3  Russia         22977474       399426         22345821        232227
  Country  Confirmed_Cases  Death_Cases  Recovered_Cases  Active_Cases
1     USA        103802702      1123836        101858395        820471
2  Brazil         37076053       699276         36124991        251786
0   India         44690738       530779         44158339          1600
49682737.0
820471
166176
  Country  Death_Rate
0   India    1.187671
1     USA    1.082665
2  Brazil    1.886058
3  Russia    1.738337
4  France    0.416829
  Country  Recovery_Rate
0   India      98.808704
1     USA      98.126921
2  Brazil      97.434835
3  Russia      97.250990
4  France      99.414502
```